ArrayList

Example 1

```
import java.util.ArrayList;

public class A_GenericNonGenericExample {
	public static void main(String[] args) {
		// Non-generic collection
		ArrayList list1 = new ArrayList();

		list1.add("Ravi");// String object
		list1.add(100);// int - Integer object
		list1.add(10.5);

		System.out.println("Non-Generic Collection:");
		System.out.println(list1);

		// Generic collection
		ArrayList<String> list2 = new ArrayList<String>();
		list2.add("Vijay");
		list2.add("Ajay");
		list2.add("Ravi");

		// Not possible
		// ArrayList<int> list3 = new ArrayList<int>();

		System.out.println("Generic Collection:");
		System.out.println(list2);

	}
}
```

Example 2

```
import java.util.*;
class B_ArrayList
{
 public static void main(String args[])
 {
        ArrayList<String> list=new ArrayList<String>();//Creating
arraylist
        list.add("Ravi");//Adding object in arraylist
        list.add("abi");
        list.add("Ram");
        list.add("Ajay");

        //Traversing list through Iterator
        Iterator iterator =list.iterator();

        while(iterator.hasNext())
        {
            System.out.println(iterator.next());
```

```
        }
        //We can also print the values just by printing object.
        System.out.println(list);
 }
}
```

Example 3

```
import java.util.*;

class C_ArrayList {
      public static void main(String args[]) {
            ArrayList<String> al = new ArrayList<String>();
            al.add("Ravi");
            al.add("Arun");
            al.add("Ravi");
            al.add("Ajay");
            // Traversing list through for-each loop
            for (String obj : al) {
                  System.out.println(obj);
                  // al.remove("Ravi"); // exception - concurrent
Modification exception
            }

      }
}
```

Example 4

```
import java.util.*;

class D_ArrayList {

      public static void main(String[] args) {

            ArrayList<String> names = new ArrayList<>();
            names.add("John");
            names.add("David");

            ArrayList<String> newNames = new ArrayList<>();
            newNames.add("Kumar");
            newNames.add("Anu");

            names.addAll(newNames);

            System.out.println(names);
      }

}
```

Example 5

```java
import java.util.ArrayList;

public class E_ArrayList {

    public static void main(String[] args) {

        ArrayList<String> names = new ArrayList<>();

        // Adding values
        names.add("Ravi");
        names.add("Priya");
        names.add("Kumar");
        names.add("Anu");

        System.out.println("Original List: " + names);

        // Add value at a particular index
        names.add(2, "Arun");
        System.out.println("After adding at index 2: " + names);

        // Remove value at a particular index
        names.remove(1);
        System.out.println("After removing index 1: " + names);

        // Remove all values
        names.clear();
        System.out.println("After removing all values: " + names);
    }
}
```

Example 6

```java
import java.util.ArrayList;

public class F_ArrayList {

    public static void main(String[] args) {

        ArrayList<String> names = new ArrayList<>();

        names.add("Ravi");
        names.add("Priya");
        names.add("Kumar");

        ArrayList<String> newNames = new ArrayList<>();

        newNames.add("Anu");
        newNames.add("Arun");

        // Add all values starting from index 1
```

```
        names.addAll(1, newNames);

        System.out.println(names);
    }
}
```

Example 7

```
import java.util.ArrayList;

public class G_ArrayList {

    public static void main(String[] args) {

        ArrayList<String> names = new ArrayList<>();

        names.add("Ravi");
        names.add("David");
        names.add("Kumar");
        names.add("Anu");

        ArrayList<String> removeNames = new ArrayList<>();

        removeNames.add("David");
        removeNames.add("Anu");

        System.out.println("Before: " + names);

        names.removeAll(removeNames);

        System.out.println("After: " + names);
    }
}
```

Example 8

```
//get method
import java.util.*;

class Employee
{
    int id;
    String name;
    String job;

    Employee(int id, String name, String job)
    {
        this.id = id;
        this.name = name;
        this.job = job;
    }
```

```
    public String toString() {
        return id + " - " + name + " - " + job;
    }
}

public class H_EmployeeDemo {
    public static void main(String[] args) {

        // Create ArrayList of Employee
        ArrayList<Employee> empList = new ArrayList<Employee>();

       //  Employee employee1 = new Employee(101, "Ravi", "Developer");
       //  empList.add(employee1);

        // Add employees
        empList.add(new Employee(101, "Ravi", "Developer"));
        empList.add(new Employee(102, "Arun", "Tester"));
        empList.add(new Employee(103, "David", "Manager"));

        System.out.println("---- Using for loop ----");
        for (int i = 0; i < empList.size(); i++) {
            System.out.println(empList.get(i));
        }

        System.out.println("\n---- Using for-each loop ----");
        for (Employee e : empList) {
            System.out.println(e);
        }

        System.out.println("\n---- Using Iterator ----");
        Iterator<Employee> itr = empList.iterator();
        while (itr.hasNext()) {
            System.out.println(itr.next());
        }
    }
}
```

Example 9

```
import java.util.*;
class I_ArrayList7
{
        public static void main(String [] args)
        {
          ArrayList<String> arrayList = new ArrayList<String>();
          System.out.println("Is ArrayList Empty: "+arrayList.isEmpty());
          arrayList.add("Ravi");
          arrayList.add("Vijay");
          arrayList.add("Ajay");
          System.out.println("After Insertion");
          System.out.println("Is ArrayList Empty: "+arrayList.isEmpty());
       }
}
```

Example 10

```
import java.util.ArrayList;

public class J_ContainsMethod {
	public static void main(String[] args) {

		ArrayList<String> names = new ArrayList<>();

		names.add("Ravi");
		names.add("Kumar");
		names.add("Anu");

		boolean result = names.contains("Anu");

		System.out.println(result);
	}
}
```

Example 11

```
import java.util.*;

class Student {
	int rollno;
	String name;
	int age;

	Student(int rollno, String name, int age) {
		this.rollno = rollno;
		this.name = name;
		this.age = age;
	}

	public String toString() {
		return rollno + " " + name + " " + age;
	}

}

class StudentArrayListMain {
	public static void main(String args[]) {

		Student student1 = new Student(104, "Sonoo", 23);
		Student student2 = new Student(101, "Ravi", 21);
		Student student3 = new Student(103, "Hanumat", 25);

		ArrayList<Student> arrayList = new ArrayList<Student>();
		arrayList.add(student1);// adding Student class object
		arrayList.add(student2);
		arrayList.add(student3);

		// Getting Iterator
		Iterator iterator = arrayList.iterator();
```

```
            while (iterator.hasNext()) {
                  System.out.println(iterator.next()); // Prints the
reference
                  // Student student=(Student)iterator.next(); //student1
                  // System.out.println(student.rollno+" "+student.name+"
"+student.age);
            }

            // System.out.println(arrayList); //Prints the reference

      }
}
```

Example 12

```
import java.util.ArrayList;
//contains method internally uses equals
class Account {
    int id;
    String name;

    Account(int id, String name) {
        this.id = id;
        this.name = name;
    }


    @Override
    public boolean equals(Object obj) {
      Account e = (Account) obj;

        return this.id == e.id &&
               this.name.equals(e.name);
    }



}

public class AccountContains {
    public static void main(String[] args) {

        ArrayList<Account> accounts = new ArrayList<>();

        Account a1 = new Account(101, "John");

        accounts.add(a1);



        //with this a2 im going to check
        Account a2 = new Account(101, "John");
```

```
        //checking the accounts collection that, whether
        //that accounts collection contains the data 101 John
        System.out.println(accounts.contains(a2));




        //ArrayList - contains - Integer class

        ArrayList<Integer> data = new ArrayList<>();
        data.add(10);//Integer object
        data.add(20);
        data.add(30);
        System.out.println("Checking");
        System.out.println(data.contains(30)); //true



        //Integer + ArrayList-correct


    }
}
```

LinkedList

Example 1

```
import java.util.*;
public class LinkedList1
{
public static void main(String args[])
{
   LinkedList<String> linkedList = new LinkedList<String>();
   linkedList.add("Ravi");
   linkedList.add("Vivek");
   linkedList.add("Ravi");
   linkedList.add("Ajay");

   Iterator<String> iterator  = linkedList.iterator();
   while(iterator .hasNext())
   {
      System.out.println(iterator.next());
   }
   for(String name:linkedList)
      {
            System.out.println(name);
      }
 }
```

```
}
```

Example 2

```
import java.util.*;
public class LinkedList2
{
 public static void main(String args[])
 {
              LinkedList<String> linkedList1=new LinkedList<String>();
            System.out.println("Initial list of elements: "+linkedList1);
            linkedList1.add("Ravi");
            linkedList1.add("Vijay");
            linkedList1.add("Ajay");
                linkedList1.add(null);
            System.out.println("After invoking add(E e) method:
"+linkedList1);
            //Adding an element at the specific position
            linkedList1.add(1, "Gaurav");
            System.out.println("After invoking add(int index, E element)
method: "+linkedList1);
            LinkedList<String> linkedList2 =new LinkedList<String>();
            linkedList2.add("Sonoo");
            linkedList2.add("Hanumat");
            //Adding second list elements to the first list
            linkedList1.addAll(linkedList2);
            System.out.println("After invoking addAll(Collection<? extends
E> c) method: "+linkedList1);
            LinkedList<String> linkedlist3 = new LinkedList<String>();
            linkedlist3.add("John");
            linkedlist3.add("Rahul");
            //Adding second list elements to the first list at specific
position
            linkedList1.addAll(1, linkedlist3);
            System.out.println("After invoking addAll(int index,
Collection<? extends E> c) method: "+linkedList1);
            //Adding an element at the first position
            linkedList1.addFirst("Lokesh");
            System.out.println("After invoking addFirst(E e) method:
"+linkedList1);
            //Adding an element at the last position
            linkedList1.addLast("Harsh");
            System.out.println("After invoking addLast(E e) method:
"+linkedList1);

    }
}
```

Example 3

```
import java.util.*;
public class LinkedList3 {

        public static void main(String [] args)
        {
```

```
            LinkedList<String> linkedList1=new LinkedList<String>();
            linkedList1.add("Ravi");
            linkedList1.add("Vijay");
            linkedList1.add("Ajay");
            linkedList1.add("Anuj");
            linkedList1.add("Gaurav");
            linkedList1.add("Harsh");
            linkedList1.add("Virat");
            linkedList1.add("Gaurav");
            linkedList1.add("Harsh");
            linkedList1.add("Amit");
            System.out.println("Initial list of elements: "+linkedList1);
          //Removing specific element from arraylist
               linkedList1.remove("Vijay");
               System.out.println("After invoking remove(object) method:
"+linkedList1);
          //Removing element on the basis of specific position
               linkedList1.remove(0);
               System.out.println("After invoking remove(index) method:
"+linkedList1);
               LinkedList<String> linkedList2=new LinkedList<String>();
               linkedList2.add("Ravi");
               linkedList2.add("Hanumat");
          // Adding new elements to arraylist
               linkedList1.addAll(linkedList2);
               System.out.println("Updated list : "+linkedList1);
          //Removing all the new elements from arraylist
               linkedList1.removeAll(linkedList2);
               System.out.println("After invoking removeAll() method:
"+linkedList1);
          //Removing first element from the list
               linkedList1.removeFirst();
               System.out.println("After invoking removeFirst() method:
"+linkedList1);
           //Removing last element from the list
               linkedList1.removeLast();
               System.out.println("After invoking removeLast() method:
"+linkedList1);
           //Removing first occurrence of element from the list
               linkedList1.removeFirstOccurrence("Gaurav");
               System.out.println("After invoking removeFirstOccurrence()
method: "+linkedList1);
           //Removing last occurrence of element from the list
               linkedList1.removeLastOccurrence("Harsh");
               System.out.println("After invoking removeLastOccurrence()
method: "+linkedList1);

               //Removing all the elements available in the list
               linkedList1.clear();
               System.out.println("After invoking clear() method:
"+linkedList1);
        }
    }
```

Example 4

```
import java.util.*;
class Book
{
	int id;
	String name,author,publisher;
	int quantity;

	public Book(int id, String name, String author, String publisher,
int quantity)
	{
		this.id = id;
		this.name = name;
		this.author = author;
		this.publisher = publisher;
		this.quantity = quantity;
	}
}


public class LinkedList4
{
	public static void main(String[] args)
	{
		List<Book> list=new LinkedList<Book>();
		Book book1 = new Book(101,"Let us C","Yashwant
Kanetkar","BPB",8);
		Book book2 = new Book(102,"Data Communications &
Networking","Forouzan","Mc Graw Hill",4);
		Book book3 = new Book(103,"Operating
System","Galvin","Wiley",6);
		//Adding Books to list
		list.add(book1);
		list.add(book2);
		list.add(book3);
		//Traversing list
		for(Book book:list)
		{
			System.out.println(book.id+" "+book.name+"
"+book.author+" "+book.publisher+" "+book.quantity);
		}
	}
}
```

HashSet

Example 1

```
import java.util.*;
class HashSet1
{
 public static void main(String args[])
 {

                HashSet<String> set=new HashSet();
            set.add("One");
            set.add("Two");
            set.add("Three");
            set.add("Four");
            set.add("Five");
                set.add("One");//removed
                for (String value : set) {
                        System.out.println(value);
                   }
 }
}
```

Example 2

```
import java.util.*;

import java.io.*;


public class HashSet2

{

        public static void main(String[] args)

        {

                Set<String>  hset = new HashSet<String>();

                hset.add("java");

                hset.add("Angular");

                hset.add("oracle");

                hset.add("oracle");

               System.out.println(hset);

                System.out.println("Angular present in the hash set  " +hset.contains("vb"));

                hset.remove("Angular");

                System.out.println(hset);
```

```
                Iterator iterator = hset.iterator();
    while (iterator.hasNext())
                {
     //String courses = (String)iterator.next();
                  //System.out.println(courses);
                 System.out.println(iterator.next());
                }
                System.out.println(hset);


                }
}
```

Example 3

```
import java.util.HashSet;
import java.util.Objects;
class Employee {
    int id;
    String name;

    Employee(int id, String name) {
        this.id = id;
        this.name = name;
    }


    @Override
    public String toString() {
        return id + " - " + name;
    }


      @Override
      public int hashCode() {
            return Objects.hash(Integer.valueOf(id), name);
      }


      @Override
      public boolean equals(Object obj) {
            if (this == obj)
```

```
            return true;
        if (obj == null)
            return false;
        if (getClass() != obj.getClass())
            return false;
        Employee other = (Employee) obj;
        return id == other.id && Objects.equals(name, other.name);
    }


}

public class HashSetEmployee {

    public static void main(String[] args) {

        HashSet<Employee> employees = new HashSet<>();

        Employee e1 = new Employee(101, "John");
        Employee e2 = new Employee(102, "Peter");
        Employee e3 = new Employee(101, "John");
        Employee e4 = new Employee(103, "Abi");

        // Duplicate employee
        employees.add(e1);
        employees.add(e2);
        employees.add(e3);
        employees.add(e4);


        for (Employee e : employees) {
            System.out.println(e);
        }


        System.out.println(e1.hashCode());
        System.out.println(e3.hashCode());

        }
}
```

Sorting array

```
public class A_SortingArray
{
        public static void main(String[] args)
        {
            int[] intArr = {5,9,1,1};
            Arrays.sort(intArr); // Sort method for array
            System.out.println("sorted Integer Array  ");
            System.out.println(Arrays.toString(intArr));
```

```
            }
}
```

Sorting Collection

```
public class A_Sorting
{
            public static void main(String[] args)
            {
                  String[] strArr = {"F", "C", "B", "Z", "E"};
                  Arrays.sort(strArr);
                  System.out.println("sorting String Array  ");
                  System.out.println(Arrays.toString(strArr));

            }
}
```

```
public class A_Sorting
{
            public static void main(String[] args)
            {
            List<Integer> integerList = new ArrayList<Integer>();
        integerList.add(13);
        integerList.add(5);
        integerList.add(7);
        integerList.add(2);
        integerList.add(1);
         Collections.sort(integerList);
            System.out.println("sorted Integer objects");
        System.out.println(integerList);
            }
}
```

Sorting user defined objects

```
import java.util.*;

class Book implements Comparable<Book> {

    int id;
    String name, author, publisher;
    int quantity;

    public Book(int id, String name, String author, String publisher, int
quantity) {
        this.id = id;
        this.name = name;
        this.author = author;
        this.publisher = publisher;
        this.quantity = quantity;
```

```
    }

    // Sorting Books based on Name in Ascending order

      public int compareTo(Book book)
      {

         return -name.compareTo(book.name);
      }


    //ascending order
    /*
    public int compareTo(Book book)
    {
      if(id>book.id){
          return -1;
      }else if(id<book.id){
          return 1;
      }else{
      return 0;
      }
    }
    */
}

public class BookComparable {

    public static void main(String[] args) {


        ArrayList<Book> list = new ArrayList<Book>();

        Book book1 = new Book(5, "Spring Boot", "Yashwant Kanetkar",
"BPB", 8);
        Book book2 = new Book(3, "Angular", "Galvin", "Wiley", 6);
        Book book3 = new Book(1, "Java", "Forouzan", "Mc Graw Hill", 4);

        list.add(book1);
        list.add(book2);
        list.add(book3);

        // Sort the ArrayList using compareTo()
        Collections.sort(list);

        for (Book book : list) {
            System.out.println(book.id + " " + book.name + " "
                    + book.author + " " + book.publisher + " " +
book.quantity);
        }

    }
}
```

```
//Sorting Books based on Name in Ascending order

/*
      public int compareTo(Book book)
      {

         return name.compareTo(book.name);
      }

*/



//Sorting Books based on Name in Decending order

/*

      public int compareTo(Book book)
      {

         return -name.compareTo(book.name);
      }

*/



//Sorting Books based on  id  in  Ascending Order

/*
public int compareTo(Book book)
{
  if(id>book.id){
      return 1;
  }else if(id<book.id){
      return -1;
  }
  else
  {
  return 0;
  }
}
*/

//Sorting Books based in id in  Descending Order

/*
public int compareTo(Book book)
{
  if(id>book.id){
```

```
      return -1;
  }else if(id<book.id){
      return 1;
  }else{
  return 0;
  }
}


*/
```